Assignment 1

1) Generative vs Discriminative model

- **Discriminative model:** learns the boundary between classes by modeling (P(Y|X)). It Predict labels from inputs.
  Example: Logistic Regression, CNN classifier.
- **Generative model:** Learns: P(X) to learn the full data distribution.
  Example: VAE, GAN.

2) CNN for chest X-ray classification

This is a **discriminative model**.
It learns how to map images → label (normal/abnormal).
It **cannot generate new X-ray images** because it only learns decision boundaries, not data distribution.

3) Meaning of learning (P(X))

Learning (P(X)) means learning which data configurations are likely.

All images compete for probability mass because:

- The total probability must sum to 1.
- If one image becomes more probable, others must become less probable.

4) Difference between (P(X)) and (P(X|Y))

- (P(X)): generate any image.
- (P(X|Y)): generate an image of a specific class.

Ex.: Generate *any face* vs generate *a smiling face*.

5) Four generative model families

Model	Computes P(X) directly?
Autoregressive models	Yes
Variational Autoencoders (VAE)	Approximate
GANs	No
Diffusion Models	No explicit direct density

6) Why high-dimensional pixel modeling is hard

Images contain thousands of pixels → extremely large combinations → curse of dimensionality → impossible to model directly.

7) Likelihood

Likelihood = how probable the observed data is under the model.
Important because training tries to **maximize likelihood**.

8) KL divergence

Measures how different two probability distributions are.
It tells us how much information is lost when approximating one distribution with another.

“How surprised would distribution Q be if reality actually followed P?”

9) Hidden variables (latent variables)

They capture hidden factors like:For faces:

- Hair color
- Age
- Smile
- Lighting

For speech:

- Accent
- Emotion
- Pitch
- Speaking speed
  They explain variations in data.

6 a) Scores = $\{58, 63, 71, 67, 66\}$

(μ_{MLE}) Mean MLE $= \dfrac{58+63+71+67+66}{5} = 65$

$\sigma^2 = \frac{1}{n} \sum (x_i - \mu)^2$

$\sum (x_i - \mu)^2 = (65-58)^2 \quad 7^2$

$+ (65-63)^2 \quad 2^2$

$+ (65-71)^2 \quad 6^2$

$+ (65-67)^2 \quad 2^2$

$+ (66-66)^2 = 94 \quad 1^2$

Variance $\sigma^2_{MLE} = 18.8$

standard deviation $\sigma_{MLE} = \sqrt{\frac{94}{5}} = 4.33$

b) MLE estimate is defined as the value of μ that maximize the log-likehood

→ Placing $\mu = 50$ placess the Λ far from the data points Making the observation Much less probable

c) log Likehood at MLE -14.3

at $\mu = 50$ is -38.7

the difference $= -14.3 - (-38.7) \Rightarrow 24.4 \rightarrow$ MLE is Much better →

$\frac{\text{MLE}}{\text{Log } \mu = 50} = e^{24.4} \rightarrow$ MLE is $e^{24.4}$ better

$-14.3 > -38.7$

less negative better log likehood better fit the model is

→ The Scores are Modeled by normal distribuation centered at 65 with spread 4.34"

(b) Why MLE has higher likelihood than μ=50

MLE chooses parameters that maximize likelihood → therefore always best fit.

(c) Log-likelihood comparison

$-14.3 > -38.7$
Large difference → $\mu=50$ fits the data **very poorly**.